College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 1

Subject: 392153 Chemistry III

Date: 28 November 2016

Year: 2016

Section: 15 -16

Time: 08.00 - 10.00

Name: ______________________ ID: ______________ Class: ____________

Instructions:

1. Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks are not allowed.
4. Dictionaries are not permitted.
5. Calculator is not permitted.
6. Electronic communication devices are not allowed in the examination room.
7. The examination has 7 pages (including this page and equation sheet), a total score of 70 points.
8. Write your Solutions and answers on the examination sheet.

You are not allowed to leave the exam room during the first one hour and 30 minutes after the beginning of the exam.

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You are not allowed to use the restroom during the exam except in case of an emergency.

Value: 70 marks Suggested Time: 120 minutes

INSTRUCTIONS: Written Response

Students will be expected to communicate your knowledge and understanding of chemical principles in a clear and logical manner. Students' steps and assumptions leading to a solution must be written in the spaces below the questions. Answers must include units where appropriate and be given to the correct number of significant figures. **For questions involving calculation and explanation, full marks will NOT be given for providing only an answer.**

1. (10 points) Consider the reaction

$$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$$

At a particular reaction time, N_2 is reacting at the rate of 0.074 M/s. Answer the following questions.

1.1 Write the rate expression of the reaction.

1.2 Calculate the rate of NH_3 forming.

1.3 Calculate the rate of H_2 reacting.

2. (10 points) The following data were collected for the reaction between A and B at 700°C.

$$2A + 2B \longrightarrow 2C + D$$

Experiment	A	B	Initial rate (M/s)
1	0.010	0.0125	0.60×10^{-6}
2	0.010	0.025	2.4×10^{-6}
3	0.0050	0.025	1.2×10^{-6}

2.1 Determine the rate law and the order of overall reaction.

2.2 Calculate the rate constant.

3. (10 points) The reaction

$$2NOBr_{(g)} \longrightarrow 2NO_{(g)} + Br_{2(g)}$$

is a <u>second order</u> reaction with respect to NOBr. The rate constant (k) = 0.810 $M^{-1}.s^{-1}$ at $10^{\circ}C$.

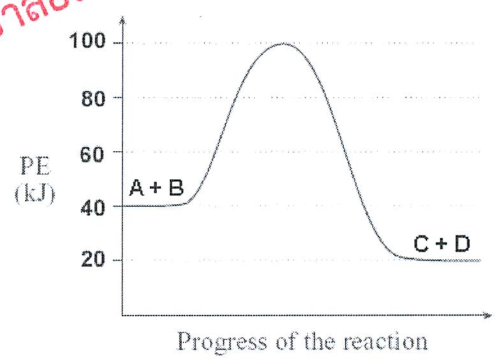
3.1 If the initial concentration of NOBr is 7.5 x 10^{-3} M, what is the concentration at 10 minutes?

3.2 Explain "half-life"

3.3 Calculate the half-life when the initial concentration of NOBr is 5.4 x 10^{-2} M.

4. (10 points) Use the concept of collision theory to explain why increasing temperatures results in an increase in reaction rate.

5. (10 points) The potential diagram below represents the energy changes in a chemical reaction. (Reactants: A and B, Products: C and D)



5.1 The heat of reaction for the forward direction is____________________kJ/mol

5.2 The energy of the activated complex is____________________kJ/mol

5.3 The activation energy (E_a) of this reaction is____________________kJ/mol

5.4 A catalyst____________(increase, decrease) the reaction rate.

5.5 <u>Draw</u> an extra potential energy line (on the diagram above) for the reaction that a catalyst is added.

6. (10 points) The activation energy of a first order reaction is 50.2 kJ/mol at 25°C. At what temperature will the rate constant double?

7. (10 points) A particular chemical reaction consists of two elementary steps, shown below. The reaction rate is indicated qualitatively (fast/slow) in each step.

Elementary step 1: $NO_{(g)} + Cl_{2(g)} \rightarrow NOCl_{2(g)}$ (slow)

Elementary step 2: $NOCl_{2(g)} + NO_{(g)} \rightarrow 2NOCl_{(g)}$ (fast)

7.1 What is the intermediate? Also, what is the overall reaction?

7.2 Identify the rate-determining step.

7.3 What would be the rate law of the overall reaction?

Equations:

$k = A \bullet \exp(-Ea/RT)$

$R = 8.314\ J/K \bullet mol$

$\ln (k_2/k_1) = -Ea/R\,((1/T_2) - (1/T_1))$

$$\ln k = -\frac{E_a}{R}\frac{1}{T} + \ln A$$

Rate Law	Concentration-Time Equation
rate = k [A]	$\ln[A] = \ln[A]_0 - kt$
rate = k	$[A] = [A]_0 - kt$
rate = k $[A]^2$	$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$